

P1251

Diagnostic information - DME (DFI) control unit

Signal to boost pressure adjuster, bank 1, faulty

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Voltage drop, at:
 - Engine start
- ♦ Power supply or ground supply faulty
- ♦ Loose contact or corrosion on connectors or lines (signal line request)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault setting condition(s)

- The electronic adjuster indicates that the signal from the DME (DFI) control unit is faulty (frequency, periods, ...).

Fault effects

- ♦ The signal from the DME (DFI) control unit is not used by the electronic adjuster.
- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 1, sluggish

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Driving (different load ranges), then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Connecting rod (ball socket) stiff/faulty (electronic adjuster for turbocharger <=> turbocharger)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) stiff/faulty
- ♦ Turbocharger stiff/faulty

Fault setting condition(s)

- The electronic adjuster indicates that the current position deviates by at least 2.5° from the required position for more than 2 s.

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster adjustment range, bank 1

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Turbocharger not adapted following replacement
- ♦ Electronic adjuster for turbocharger has no or incorrect basic setting/adaptation
- ♦ Electronic adjuster for turbocharger stiff/faulty
- ♦ Connecting rod broken/disengaged/blocked (electronic adjuster for turbocharger - turbocharger)
- ♦ Turbocharger sluggish/jammed

Fault setting condition(s)

- The electronic adjuster indicates that the current adjustment range of the boost pressure adjuster deviates from the adjustment range taught during the teaching process (is exceeded or not reached).

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1254

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster temperature, bank 1

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Heat shield damaged
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) - exhaust system, distance too low

Fault setting condition(s)

- Boost pressure adjuster (electronic adjuster for turbocharger) temperature > 135 °C (275 °F)

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Engine torque is reduced

Diagnostic information - DME (DFI) control unit

Signal to boost pressure adjuster, bank 2, faulty

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Voltage drop, at:
 - Engine start
- ♦ Power supply or ground supply faulty
- ♦ Loose contact or corrosion on connectors or lines (signal line request)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

Fault setting condition(s)

- The electronic adjuster indicates that the signal from the DME (DFI) control unit is faulty (frequency, periods, ...).

Fault effects

- ♦ The signal from the DME (DFI) control unit is not used by the electronic adjuster.
- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 2, sluggish

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Driving (different load ranges), then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Connecting rod (ball socket) stiff/faulty (electronic adjuster for turbocharger <=> turbocharger)
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) stiff/faulty
- ♦ Turbocharger stiff/faulty

Fault setting condition(s)

- The electronic adjuster indicates that the current position deviates by at least 2.5° from the required position for more than 2 s.

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster adjustment range, bank 2

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Turbocharger not adapted following replacement
- ♦ Electronic adjuster for turbocharger has no or incorrect basic setting/adaptation
- ♦ Electronic adjuster for turbocharger stiff/faulty
- ♦ Connecting rod broken/disengaged/blocked (electronic adjuster for turbocharger - turbocharger)
- ♦ Turbocharger sluggish/jammed

Fault setting condition(s)

- The electronic adjuster indicates that the current adjustment range of the boost pressure adjuster deviates from the adjustment range taught during the teaching process (is exceeded or not reached).

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1258

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster temperature, bank 2

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Heat shield damaged
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) - exhaust system, distance too low

Fault setting condition(s)

- Boost pressure adjuster (electronic adjuster for turbocharger) temperature > 135 °C (275 °F)

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Engine torque is reduced

P1261

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 1, not adapted

(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Bank 1 boost pressure adjuster (electronic adjuster for turbocharger) not adapted

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1262

Diagnostic information - DME (DFI) control unit

Boost pressure adjuster, bank 2, not adapted
(Electronic adjuster for turbocharger)

Diagnostic conditions

- Ignition on, then:
 - Switch off ignition
 - Switch on ignition

Possible fault causes

- ♦ Bank 1 boost pressure adjuster (electronic adjuster for turbocharger) not adapted

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

P1265

Diagnostic information - DME (DFI) control unit

Crash signal

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the wiring for the signal line (airbag triggering unit)

P1280

Diagnostic information - DME (DFI) control unit

Fuel pump control unit

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Fuel pump control unit faulty

P1280

Diagnostic information - DME (DFI) control unit

Fuel pump control unit

- Only for all-wheel drive vehicles

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Fuel pump control unit implausible

Fault effects

- ♦ The control unit attempts to bring the fuel pump to the setpoint value.

P1281

Diagnostic information - DME (DFI) control unit

Electric fuel pump: electrical fault

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in wiring for fuel pump control unit ⇒ Electric fuel pump
- ♦ Open circuit in ground line for electric fuel pump
- ♦ Electric fuel pump faulty

P1281

Diagnostic information - DME (DFI) control unit

Fuel pump control unit: overtemperature

- Only for all-wheel drive vehicles

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Overloading of fuel pump control unit

Fault effects

- ♦ The control unit attempts to bring the fuel pump to the setpoint value.

P1282

Diagnostic information - DME (DFI) control unit

Electric fuel pump: sluggish

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Foreign body in electric fuel pump
- ♦ Fuel line blocked
- ♦ Electric fuel pump faulty

Fault setting condition(s)

- Current draw max. 30% > setpoint value

Fault effects

- ♦ The control unit attempts to bring the fuel pump to the setpoint value.

P1283

Diagnostic information - DME (DFI) control unit

Electric fuel pump: electrical fault

- Only for rear-wheel drive vehicles

Diagnostic conditions

- Engine start (electric fuel pump activated)

Possible fault causes

- ♦ Supply voltage for fuel pump control unit too low
- ♦ Supply voltage for fuel pump control unit too high
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- Supply voltage for fuel pump control unit < 3 V
- or
- Supply voltage for fuel pump control unit > 15 V for > 1 hour(s)
- or
- Supply voltage for fuel pump control unit > 17 V for > 1 minute(s)

P1283

Diagnostic information - DME (DFI) control unit

Electric fuel pump: blocked

- Only for all-wheel drive vehicles

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Foreign body in electric fuel pump
- ♦ Electric fuel pump faulty

Fault setting condition(s)

- Current draw at least 30% > setpoint value

Fault effects

- ♦ The control unit attempts to bring the fuel pump to the setpoint value.

P1284

Diagnostic information - DME (DFI) control unit

Electric fuel pump: electrical fault

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Vehicle voltage too low due to discharged or faulty starter battery
 - ⓘ See Extended fault memory, >[40-Control Module Voltage](#)<
- ♦ Short circuit in lines for fuel pump control unit ⇒ Electric fuel pump
- ♦ Electric fuel pump faulty

Fault effects

- ♦ Electric fuel pump switched off
- ♦ Activation and testing of electric fuel pump after 50 ... 150 millisecond(s)

P1285

Diagnostic information - DME (DFI) control unit

Electric fuel pump: electrical fault

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Fuel tank empty
 - ⓘ See Extended fault memory, >[40-Fuel Level Input](#)<
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Open circuit in lines for fuel pump control unit ⇒ Electric fuel pump
- ♦ Open circuit in ground line for electric fuel pump
- ♦ Electric fuel pump faulty

Fault effects

- ♦ Electric fuel pump switched off
- ♦ Activation and test of electric fuel pump after 50 ... 150 millisecond(s)

P1286

Diagnostic information - DME (DFI) control unit

Speed deviation

Diagnostic conditions

- Switch on ignition or engine start (electric fuel pump activated)

Possible fault causes

- ♦ Supply voltage for fuel pump control unit too low
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Leak in fuel line
- ♦ Fuel line blocked
- ♦ Foreign body in electric fuel pump
- ♦ Electric fuel pump sluggish

Fault setting condition(s)

- Speed of electric fuel pump deviates from setpoint speed, at:
 - ♦ Supply voltage for fuel pump control unit < 12.5 V
 - ♦ Fuel counter-pressure too high
 - ♦ Fuel counter-pressure too low
 - ♦ Electric fuel pump sluggish

Fault effects

- ♦ The control unit attempts to bring the fuel pump to the setpoint value.

P1300

Diagnostic information - DME (DFI)

Intake valve lift adjustment solenoid valve, bank 1

Diagnostic conditions

- Short test: Intake valve lift switchover valves

or

- Drive for longer with
 - Engine at operating temperature
 - Engine speed 1,200 to 5,000 rpm
 - Several and longer lasting switch-overs to the large lift [\(A220\)](#), for heating the valve and reliable detection

Possible fault causes

- ♦ Intake valve lift adjustment solenoid valve on bank 1 mechanically sluggish or blocked

Fault setting condition(s)

- Following activation of the solenoid valve, the mechanical movement of the valve is detected based on the flowback (induction) in the coil. If there is insufficient or no flowback, a fault is entered.

P1301

Diagnostic information - DME (DFI)

Intake valve lift adjustment solenoid valve, bank 2

Diagnostic conditions

- Short test: Intake valve lift switchover valves

or

- Drive for longer with
 - Engine at operating temperature
 - Engine speed 1,200 to 5,000 rpm
 - Several and longer lasting switch-overs to the large lift [\(A220\)](#), for heating the valve and reliable detection

Possible fault causes

- ♦ Intake valve lift adjustment solenoid valve on bank 2 mechanically sluggish or blocked

Fault setting condition(s)

- Following activation of the solenoid valve, the mechanical movement of the valve is detected based on the flowback (induction) in the coil. If there is insufficient or no flowback, a fault is entered.

P1302

Diagnostic information - DME (DFI)

Exhaust valve lift adjustment solenoid valve, bank 1

Diagnostic conditions

- Short test: Exhaust valve lift switchover valves

or

- Drive for longer with
 - Engine at operating temperature
 - Engine speed 1,200 to 5,000 rpm
 - Several and longer lasting switch-overs to the large lift [\(A220\)](#), for heating the valve and reliable detection

Possible fault causes

- ♦ Exhaust valve lift adjustment solenoid valve on bank 1 mechanically sluggish or blocked

Fault setting condition(s)

- Following activation of the solenoid valve, the mechanical movement of the valve is detected based on the flowback (induction) in the coil. If there is insufficient or no flowback, a fault is entered.

P1303

Diagnostic information - DME (DFI)

Exhaust valve lift adjustment solenoid valve, bank 1

Diagnostic conditions

- Short test: Exhaust valve lift switchover valves

or

- Drive for longer with
 - Engine at operating temperature
 - Engine speed 1,200 to 5,000 rpm
 - Several and longer lasting switch-overs to the large lift [\(A220\)](#), for heating the valve and reliable detection

Possible fault causes

- ♦ Exhaust valve lift adjustment solenoid valve on bank 1 mechanically sluggish or blocked

Fault setting condition(s)

- Following activation of the solenoid valve, the mechanical movement of the valve is detected based on the flowback (induction) in the coil. If there is insufficient or no flowback, a fault is entered.

Diagnostic information - DME (DFI) control unit

Valve lift control, bank 1

Diagnostic conditions

- Perform a short test

or

- Drive with engine well heated at operating temperature with slight acceleration :

1. Large lift (actual value A220):

- Time since engine start > 5 minute(s)
- Engine speed 1,250 ... 2,100 rpm (actual value A020)
- Mass air flow 300 ... 600 mg/stroke (actual value H010)

2. Small lift (actual value A220):

- Time since engine start > 5 minute(s)
- Engine speed 1,250 ... 1,600 rpm (actual value A020)
- Mass air flow 180 ... 350 mg/stroke (actual value H010)

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Engine oil pressure too high or too low
- ♦ Loose contact or corrosion on connectors or lines (power supply for solenoid valve for hydraulic valve lift control)

Fault setting condition(s)

- 6-cylinder engines: Based on the oscillation behavior of the camshafts, the diagnostic function detects that all the valves have either not switched to the other lift or have not switched quickly enough during valve lift switchover.
- 4-cylinder engines: Based on the lambda controller behavior, the diagnostic function detects that all the valves have either not switched to the other lift or have not switched quickly enough during valve lift switchover.

Diagnostic information - DME (DFI) control unit

Valve lift control, bank 2

Diagnostic conditions

- Perform a short test

or

- Drive with engine well heated at operating temperature with slight acceleration :

1. Large lift (actual value A220):

- Time since engine start > 5 minute(s)
- Engine speed 1,250 ... 2,100 rpm (actual value A020)
- Mass air flow 300 ... 600 mg/stroke (actual value H010)

2. Small lift (actual value A220):

- Time since engine start > 5 minute(s)
- Engine speed 1,250 ... 1,600 rpm (actual value A020)
- Mass air flow 180 ... 350 mg/stroke (actual value H010)

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Engine oil pressure too high or too low
- ♦ Loose contact or corrosion on connectors or lines (power supply for solenoid valve for hydraulic valve lift control)

Fault setting condition(s)

- 6-cylinder engines: Based on the oscillation behavior of the camshafts, the diagnostic function detects that all the valves have either not switched to the other lift or have not switched quickly enough during valve lift switchover.
- 4-cylinder engines: Based on the lambda controller behavior, the diagnostic function detects that all the valves have either not switched to the other lift or have not switched quickly enough during valve lift switchover.

P1359

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

P1360

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

P1361

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

P1362

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

P1363

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

P1364

Diagnostic information - DME (DFI)

Valve lift control, cylinder X

P1359 = Cylinder 1

P1361 = Cylinder 2

P1363 = Cylinder 3

P1362 = Cylinder 4

P1364 = Cylinder 5

P1360 = Cylinder 6

Diagnostic conditions

- Short test: Intake valve lift, individual cylinder

Possible fault causes

- ♦ Intake valve tappet of the cylinder specified in the fault text is faulty

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder, calculated from the expected and actual crankshaft acceleration due to combustion. A comparison of the values in the large and small lift determines if an intake valve has not switched to the other lift or has not switched quickly enough.

Diagnostic information - DME (DFI) control unit

Valve lift control, bank 1

Diagnostic conditions

- Perform a short test

or

- Driving:
 - Engine oil temperature < 111 °C (232 °F)
 - Large lift:
 - Engine speed 2,000 ... 5,000 rpm
 - Mass air flow > 700 ... > 300 mg/stroke (depending on engine speed)
- Small lift:
 - Engine speed 1,500 ... 3,000 rpm
 - Mass air flow > 500 ... > 100 mg/stroke (depending on engine speed)

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Oil supply too low
 - ⇒ (e.g. dirty (oil circuit), ...)
- ♦ Large differences in exhaust backpressure between banks (e.g. due to constricted or blocked passage of an exhaust branch).
- ♦ Wrong timing, one bank (intake camshaft or exhaust camshaft)
- ♦ Solenoid valve for hydraulic valve lift control faulty
 - ⇒ (mechanical fault or electrical fault in circuit)
- ♦ 1 oxygen sensor aged or faulty
- For engines with VTG turbocharger:
 - ♦ Actuating rod for boost pressure adjuster disengaged

Fault setting condition(s)

- Based on the lambda controllers, the diagnostic function detects that all the valves on this bank have either not switched to the other lift or have not switched quickly enough during valve lift switchover.

Diagnostic information - DME (DFI) control unit

Valve lift control, bank 2

Diagnostic conditions

- Perform a short test

or

- Driving:
 - Engine oil temperature < 111 °C (232 °F)
 - Large lift:
 - Engine speed 2,000 ... 5,000 rpm
 - Mass air flow > 700 ... > 300 mg/stroke (depending on engine speed)
- Small lift:
 - Engine speed 1,500 ... 3,000 rpm
 - Mass air flow > 500 ... > 100 mg/stroke (depending on engine speed)

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Oil supply too low
 - ⇒ (e.g. dirty (oil circuit), ...)
- ♦ Large differences in exhaust backpressure between banks (e.g. due to constricted or blocked passage of an exhaust branch).
- ♦ Wrong timing, one bank (intake camshaft or exhaust camshaft)
- ♦ Solenoid valve for hydraulic valve lift control faulty
 - ⇒ (mechanical fault or electrical fault in circuit)
- ♦ 1 oxygen sensor aged or faulty
- For engines with VTG turbocharger:
 - ♦ Actuating rod for boost pressure adjuster disengaged

Fault setting condition(s)

- Based on the lambda controllers, the diagnostic function detects that all the valves on this bank have either not switched to the other lift or have not switched quickly enough during valve lift switchover.

P13A3

Diagnostic information - DME (DFI) control unit

Misfiring

INFORMATION 1

The entry **P13A3** (Misfiring that is damaging to catalytic converter with cylinder deactivation) in the fault memory is for information purposes only.

♦ It is stored in order to track the behavior of the vehicle.

INFORMATION 2

The entry **P0313** (Misfiring when fuel tank level is low) in the fault memory provides additional information about the faults (P0300 ... P0306), which are also entered and indicate the cylinders on which misfiring has occurred.

► In this case, check whether misfiring may have been caused only as a result of the low fuel tank level .

Diagnostic conditions

- Engine temperature > -20 °C (-4 °F)
- Idle speed up to the red area on the tachometer
- No overrun phase(s)
- None of the following faults stored: P0335, P0336, P0371, P0373

Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bad ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked
J Catalytic converter faulty

J Leak in intake-air distributor
J Foreign bodies in intake-air distributor
J Camshaft adjustment fault
J Timing wrong
J Valve lift control fault
J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)
J Mechanical fault in valve drive
J Insufficient compression
(e.g. cylinder liner damaged, valve seat ring faulty, ...)
I ONLY for manual transmission:
J Clutch and flywheel damaged as a result of thermal overload

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
- i** It is possible that only one cylinder (or the cylinders of one bank) has/have exceeded the uneven running threshold for a fault entry. Other cylinders may be just below the level which would result in a fault entry.
⇒ See actual values [M005](#) to [M060](#)

****i** INFORMATION 3**

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure

increase that is too slow, or a rapid drop in pressure when the fuel injectors open.

- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P13EA

Diagnostic information - DME (DFI) control unit

Ignition timing angle after cold start

INFORMATION

This fault entry is used for information purposes only; the actual cause of the fault is in another area.

♦ Significant engine running problems force the engine control system to leave the permitted ignition timing range in order to keep the engine running.

Diagnostic conditions

- Cold start
- Then driving

Diagnostic information - DME (DFI) control unit

Vacuum system leaking - cooling system fault

Diagnostic conditions

- Perform system test on vacuum system

OR complete the following two steps one after the other:

1. Establish the following conditions three times one after the other:

- Vehicle is stationary:
 - Engine oil temperature (T030) 30 to 110 °C (86 to 230 °F)
 - Engine idling
 - Forcefully apply the brake several times (brake pressure (A255) greater than 12 bar) until the vacuum (A250) increases to over 500 hPa
 - Then do not actuate the brake for more than 40 seconds
 - Then drive a short distance at a speed of over 2 km/h (1 mph), maintaining an engine speed of over 1,500 rpm

2. Then establish the following conditions three times one after the other:

- Actuate the brake at least once while driving (brake pressure (A255) greater than 5.5 bar) - the vacuum (A250) must increase to over 200 hPa
- Then drive at as constant an engine speed (A020) as possible between 1,000 ... 6,000 rpm for more than 60 seconds

 If vacuum-actuated valves are switched during the 60 seconds of constant driving, the diagnostic process will be interrupted and will start from the very beginning again after the gear change so that the time that must be driven in order to complete diagnostics can be much longer!

Possible fault causes

- ♦ Vacuum system leaking (see  **INFORMATION**)
- ♦ Brake light switch faulty
- ♦ Brake light switch aligned incorrectly at brake pedal
- ♦ Contact resistance on plug connection to the brake vacuum sensor
- ♦ Brake vacuum sensor faulty

Fault effects

1. The message [\[Cooling system fault\]](#) appears in the instrument cluster if a leak is detected in the vacuum system.

Reason: The cooling system with thermal management also includes vacuum-actuated shutoff valves.

2. If there is no vacuum, the vacuum-actuated flaps and valves can no longer be closed, which can lead to malfunctions.

Noticeable effects can include:

- ♦ lack of torque
- ♦ no heating power